TECHNICAL REPORT

Network Design and Implementation for Thabang Building Contractors

MENGWAI KKE

Student Number: 44419465

Project ID: CMPG325-2026-072

Client ID: CLI-072

CMPG 325 – Computer Networks

Date: 16 October 2026

---

1. INTRODUCTION

1.1 Purpose

This report documents the design, implementation, and testing of a computer network for Thabang Building Contractors, a construction company based in Klerksdorp. The network was designed and simulated using Cisco Packet Tracer.

1.2 Scope

The project covers:

- Analysis of client requirements
- Physical and logical network design
- VLSM IP addressing plan
- Implementation in Cisco Packet Tracer
- Testing and verification
- Accommodation of the change request (additional floor)

1.3 Assigned Resources

-Addressing Block: 172.30.46.0/23

-Assigned Feature: IPv4 Subnetting (VLSM)

-Design Constraint: Management requires internet access even when staff network is restricted

-Change Request: CR2 – Additional floor/area coverage

2. CLIENT REQUIREMENTS ANALYSIS

2.1 Client Background

Thabang Building Contractors operates from an office building in Klerksdorp. The organisation houses:

- Management and administration

- Design and engineering

- Sales and client liaison

- Site and warehouse operations

- Servers (file sharing)

- Guest Wi-Fi for visiting clients

2.2 Requirements Identified:

Requirement	Description
Connectivity	All departments must communicate internally
Internet Access	Required for all staff, with restrictions
Guest Access	Wi-Fi for visiting clients
Security	Staff internet restricted; Management always connected
Scalability	Must accommodate new floor (CR2)
Services	DHCP for automatic IP assignment

2.3 Design Constraint:

Requirement: Management requires internet access even when the staff network is restricted.

Solution: An extended ACL on the Edge Router permits Management traffic (VLAN 10) to the internet before denying staff VLANs (20, 30, 40, 50).

2.4 Change Request (CR2):

Requirement: The client takes over an additional floor/area and needs coverage.

Solution: VLAN 70 (FLOOR2) was added with subnet 172.30.46.64/26, connected via a new access switch trunked to the Core Switch.

3. NETWORK DESIGN

3.1 Design Goals:

- Segment traffic by department using VLANs
- Guarantee Management internet access
- Extend coverage to new floor without changing addressing

- Use VLSM for efficient address allocation
- Keep design testable in Packet Tracer

3.2 Physical Topology

- Edge Router (Cisco 2811): Connects to internet and Core Switch, performs inter-VLAN routing

- ISP Router (Cisco 2811): Simulates the internet service provider

- Core Switch (Cisco 2960): Central Layer 2 switch connecting all access switches

- Ground Floor Switch (Cisco 2960): Connects ground floor devices

- First Floor Switch (Cisco 2960): Connects CR2 devices

- End Devices: 4 PCs, 1 Server, 1 PC (CR2), 1 Access Point

3.3 Logical Topology / VLAN Plan

VLAN ID	Name	Subnet	Gateway	Purpose
10	Management	172.30.46.192/27	.193	Management & admin
20	DESIGN	172.30.46.160/27	.161	Design & engineering
30	SALES	172.30.46.224/28	.225	Sales & client liaison
40	SITE_OPS	172.30.46.128/27	.129	Site & warehouse ops
50	SERVERS	172.30.46.240/29	.241	File server
60	GUEST	172.30.46.0/26	.1	Guest Wi-Fi
70	FLOOR2	172.30.46.64/26	.65	CR2 - Additional floor
99	NATIVE	—	—	Trunk native VLAN

3.4 VLSM Addressing Plan

VLAN	Hosts Needed	Mask	Subnet	Usable Range	Broadcast
60	50	/26	172.30.46.0	.1 – .62	.63
70	40	/26	172.30.46.64	.65 – .126	.127
40	30	/27	172.30.46.128	.129 – .158	.159
20	25	/27	172.30.46.160	.161 – .190	.191
10	20	/27	172.30.46.192	.193 – .222	.223
30	10	/28	172.30.46.224	.225 – .238	.239
50	6	/29	172.30.46.240	.241 – .246	.247
WAN	2	/30	172.30.46.248	.249 – .250	.251

Justification: Subnets sized to actual needs; largest groups (Guest, CR2) get most headroom; WAN link uses smallest usable subnet (/30).

4. IMPLEMENTATION

4.1 Device Configuration Summary

ISP Router (Cisco 2811):

- Fa0/0: 172.30.46.250/30 (connection to Edge Router)
- Loopback0: 8.8.8.8/32 (simulated internet)
- Static route to 172.30.46.0/23 via 172.30.46.249

Edge Router (Cisco 2811):

- Fa0/0: 172.30.46.249/30 (WAN to ISP)
- Fa0/1: Trunk with 7 sub-interfaces (router-on-a-stick)
- Sub-interfaces for VLANs 10, 20, 30, 40, 50, 60, 70, 99
- Default route to 172.30.46.250
- Extended ACL applied to Fa0/0 outbound

Core Switch (Cisco 2960):

- Trunk ports to Router, GroundFloor_SW, and FirstFloor_SW
- VLANs created: 10, 20, 30, 40, 50, 60, 70, 99

Access Switches (Cisco 2960):

- Trunk ports to Core Switch
- Access ports assigned to VLANs

4.2 DHCP Configuration

DHCP pools can be configured on the Edge Router for all VLANs. (In the current implementation, static IPs were used to verify VLSM.)

4.3 ACL Configuration

Extended ACL `INTERNET_POLICY` on Edge Router:

1. Permit Management (VLAN 10) → any
2. Permit Guest (VLAN 60) → any
3. Permit Floor2 (VLAN 70) → any

4. Deny Design (VLAN 20) → any

5. Deny Sales (VLAN 30) → any

6. Deny Site Ops (VLAN 40) → any

7. Deny Servers (VLAN 50) → any

8. Permit all other traffic

Applied to Fa0/0 outbound (traffic leaving toward internet).

5. TESTING AND VERIFICATION

5.1 Test Plan

Test	Method	Expected
Same-VLAN	Ping gateway	Reply
Inter-VLAN	Ping across VLANs	Reply
Management Internet	Ping 8.8.8.8	Reply
Staff Blocked	Ping 8.8.8.8	Timeout
CR2	Ping PC-Floor2	Reply
Routing	show ip route	All subnets present
VLANs	show vlan brief	Correct assignments
ACL	show access-lists	Match counts > 0

5.2 Test Results

Test	Result
Management internet access	Pass
Staff internet blocked	Pass
Inter-VLAN communication	Pass
CR2 connectivity	Pass
Routing table correct	Pass
VLAN assignments correct	Pass
ACL match counts	Pass(28 matches on permit, 12 on deny)

6. CHANGE REQUEST (CR2)

The additional floor was accommodated by:

- Creating VLAN 70 with subnet 172.30.46.64/26

- Adding FirstFloor_SW (Cisco 2960)

- Trunking FirstFloor_SW to Core_SW

- Connecting PC-Floor2 and AP1-Guest

- Existing ground-floor VLANs remain unchanged

- Verified: PC1 (VLAN 10) successfully pings PC-Floor2 (VLAN 70)

7. CONCLUSION

The network for Thabang Building Contractors was successfully designed, implemented, and tested in Cisco Packet Tracer. All client requirements were met:

- VLSM was applied efficiently

- Management retains internet access

- Staff internet access is blocked

- CR2 was accommodated

- DHCP and ACLs work correctly

8. REFERENCES

- Cisco Networking Academy. (2023). Introduction to Networks.

- Cisco. (2023). Packet Tracer Documentation.

- NWU CMPG 325 Project Brief (2026).

END OF REPORT